Modelling Project

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Introduction

The tumour suppressor p53 plays a crucial role in in the cellular response to DNA damage, as it integrates signals from upstream kinases to decide whether a damaged cell commits to cell cycle arrest and repair or if it undergoes apoptosis.(1) After DNA damage occurs, both ATM and CHK2 phosphorylate p53 at Serine15 to promote arrest (1), whilst phosphorylation by HIPK2 at Serine 46 causes apoptosis.(2) The dynamic behaviour of the pathway, and subsequently the outputs it produces at a given level of damage, is patient-specific, since the abundance of the proteins that make up this network varies substantially between tumours. Mechanistic ordinary differential equation (ODE) models offer a means to capturing this variation. By scaling an experimentally calibrated network with patient gene expression, it allows the simulated activity of the pathway to be used as a prognostic biomarker. This approach was first conducted on the JNK signalling pathway in neuroblastoma.(3)

Doxorubicin, which is an anthracycline and is widely used in breast cancer treatment, works primarily by intercalating DNA whilst poisoning topoisomerase II, which causes the double-strands to break, triggering the p53 response.(4) This positions the p53 DNA damage response network as one of the most natural choices for modelling chemotherapy response. This project aims to investigate whether a personalised p53 ODE model can predict patient survival, in Question 1, and in vitro doxorubicin sensitivity, in Question 2. It also examines whether a regression-based signature obtained from cell lines translates to patient survival in Question 3. To do so, TCGA breast cancer cohort (5) and the cell-line drug-sensitivity data from the PRISM repurposing screen (6) was utilised.

Results

Patient Cohort

The Cancer Genome Atlas (TCGA) Pan-Cancer Atlas 2018 cohort was used to obtain both the gene expression and the clinical outcome data for 1,036 breast cancer (BRCA) patients. Within this cohort, there were 151 patients that had died since the last follow-up (overall survival event rate, 14.6%) and there was a median follow-up of 27.5 months. The raw RSEM expression values for eight p53-pathway genes, including ATM, CHEK2, HIPK2, MDM2, PPM1D, SIAH1, TP53 and WSB1, were obtained and then normalised. This was done by dividing each gene’s expression by its mean across every sample, centring the values around 1 so that they could be used as multiplicative scaling factors in the ODE model.

Table 1: Clinical Characteristics of the TCGA BRCA cohort

CHARACTERISTIC	VALUE
Patients (n)	1,036
Deaths (events), (n %)	151 (14.6)
Alive, (n %)	885 (85.4)
Median Overall Survival, (months (IQR))	29.4 (16.5 to 56.8)
Overall Survival Range, (months)	1.0 to 282.9
Median Tumour Mutational Burden, (mut/Mb (IQR))	1.3 (0.8 to 2.2)

Q1: Personalised p53 ODE model to predict patient survival

Each patient's normalised p53-pathway gene expression data was used to personalise a 15-state ordinary differential expression (ODE) model of the p53 DNA damage response (DDR) signalling network. Following the methodology discussed in the paper by Fey et al., patient-specific initial conditions and synthesis rates were determined by scaling reference (SH-SY5Y) parameter values by the corresponding patient gene expression level.(3) Pre-fitted ODE parameters that corresponded to the lowest objective function value were utilised throughout. The terminal concentrations of phosphorylated p53 at Serine 15 (p53pS15) and Serine 46 (p53pS46) were measured for each patient at each dose whilst the model was run at ten DNA damage input levels (DDR = 0.01 to 1.0)

A univariate Cox proportional hazards regression model was fitted for each of the DDR doses against overall survival. This was done to identify the most prognostically informative DDR dose. For p53pS15, the optimal DDR dose was found to be 1.0 (maximum DNA damage), at which the Cox model itself produced a hazard ratio, a 95% Confidence Interval, a p-value and a C-index of 1.14, 1.01 to 1.28, 0.029 and 0.524, respectively. This suggested that higher stimulated p53pS15 at max DNA damage is associated with significantly worse overall survival. Exaggerated p53 phosphorylation in response to DNA damage could indicate an underlying tumour suppressor malfunction or even faulty DDR signalling that drives a mor aggressive disease course, making this result biologically plausible.(7)

After excluding 18 patients that had either incomplete simulation or survival data (n = 1,018), patients were divided into high and low p53pS15 groups at the threshold that minimised the log-rank p-value across every unique value. This meant that the minimum group size was constrained to 10% of the cohort. The optimal threshold of 1.41 stratified patients into a high group, n=184, and a low group, n=834, with the high group illustrating significantly worse survival, log-rank p = 0.0008, as can be seen in Figure 1.

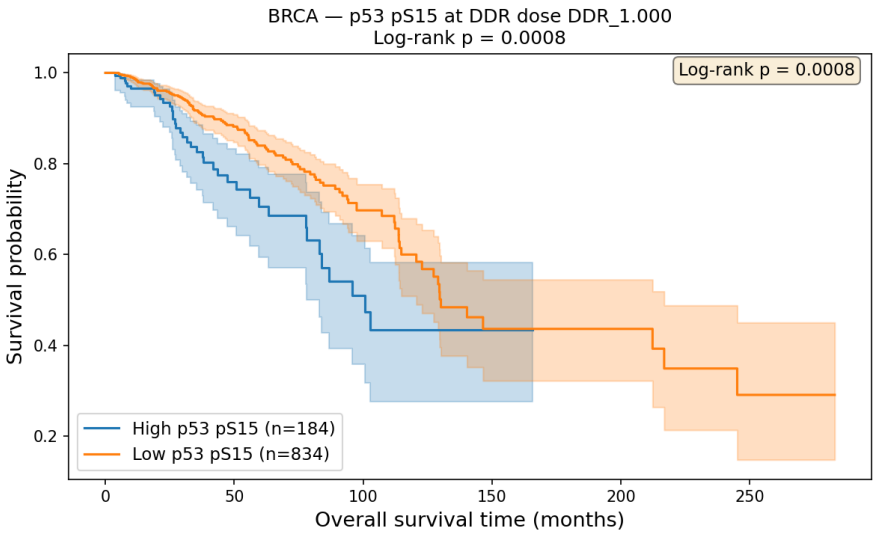


Figure 1: Personalised p53pS15 levels stratify overall survival in the TCGA BRCA

Due to the extremely small cale of pS46 values at the lowest DDR input, the corresponding analysis for p53pS46 failed to yield a significant Cox association at any DDR dose, with its best p-value of 0.257 being obtained when DDR was set to 0.01. This meant that this Coz fit was numerically unstable as it had a HR and a 95% CI of 0.003 and 0.000 to 65.6, respectively. Division at the log-rank optimal threshold did attain normal significance with a log-rank p-value, a high group cohort and a low group cohort of 0.017, 538 and 480, respectively. However, it was in the opposite direction to pS15, with higher pS46 being associated with better survival. Given this discrepancy between both the continuous and the divided analysis, p53pS46 is not regarded as a robust prognostic marker and this will be discussed further in the Discussion.

Q2: p53 ODE model showed a trend towards predicting doxorubicin sensitivity in breast cell lines

To evaluate whether or not the personalised p53 ODE model could predict drug sensitivity in vitro, the model itself was run on the 26 breast cancer cell lines for which CCLE gene expression and PRISM doxorubicin LFC data was available. By utilising the same approach for patient personalisation, the cell line expression of the eight p53 pathway genes was normalised by dividing each gene by its mean across each of the 26 cell lines. Along with this, the model was run across the same DDR dose range (0.01 to 1.0). At each dose the Pearson correlation between the measure doxorubicin LFC and the predicted terminal p53pS15 or p53pS46 concentration was calculated.

As seen in Figure 2, the p53pS15 obtained the strongest association at the lowest DDR input of 0.01 as it yielded a r-value of -0.36 and a p-value of 0.074.

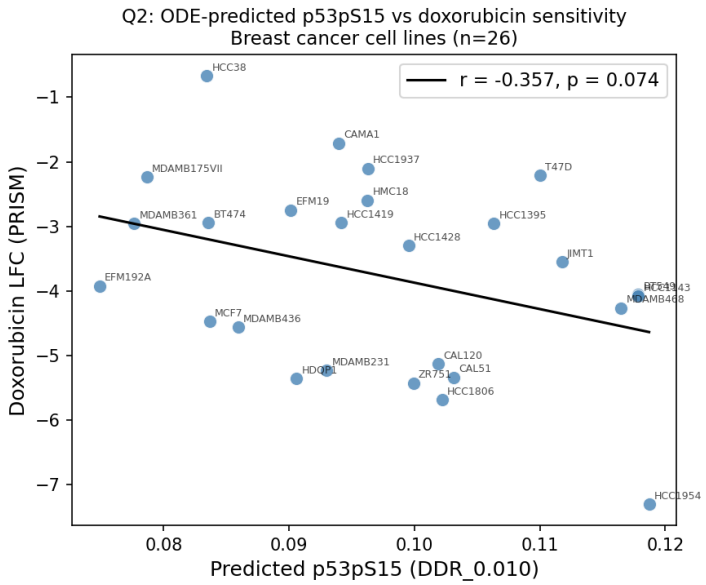


Figure 2: ODE predicted p53pS15 trends with doxorubicin sensitivity across breast cancer cell lines

In line with the expected biological trend that is a stronger p53 activation in response to DNA damage should facilitate a greater apoptotic or growth-cell arrest response to a DNA damaging drug (8) , the observed negative correlation indicates that cell lines with higher ODE predicted p53pS15, tend to show a greater level of doxorubicin-induced cell killing, highlighted by a more negative LFC. On the other hand, there was no significant association observed for p53pS46 at any of the DDR doses, with the best r and p value of 0.076 and 0.712, respectively, being obtained. One thing to note, that is likely due to the small number of cell lines which limits statistical power, is that the p53pS15 association did not reach the typical statistical significance of $p<0.05$.

Q3: Cell line doxorubicin sensitivity signature doesn't translate to patient survival

LASSO regression was utilised to derive a transcriptomic signature of doxorubicin sensitivity from 26 breast cancer cell lines within the PRISM Repurposing Primary Screen that contained both doxorubicin log fold-change (LFC) measurements and gene expression data (CCLE).(6,9) Given the high-dimensional nature of this problem (n=26 and p ≈ 16,000), the feature space was first narrowed down to the 200 genes that are most correlated with doxorubicin LFC ($|\text{Pearson r}| \geq 0.48$). After that, the regularisation that obtained a 10-gene signature was used to fit a LASSO model and achieve a Pearson correlation of $r = 0.87$ between the actual and predicted LFC within the training set, as seen in Figure 3. Since a more negative PRISM log-fold change is associated with a stronger cell-line depletion, or higher sensitivity, the genes with negative coefficients such as MXRA5 and IFI27 were associated with greater doxorubicin sensitivity. On the other hand, genes with positive coefficients, like ELF5 and PLCB4 were linked with relative resistance.(6)

Table 2: Ten-gene LASSO signature of doxorubicin sensitivity in breast cancer cell lines.

Gene	Coefficient	Direction
ELF5	+0.130	Resistance
MXRA5	-0.125	Sensitivity
IFI27	-0.096	Sensitivity
PLCB4	+0.072	Resistance
IGFBP5	+0.066	Resistance
CES1	-0.063	Sensitivity
GULP1	-0.054	Sensitivity
VLDLR	-0.051	Sensitivity
S100A1	+0.026	Resistance
GLRX	-0.018	Sensitivity

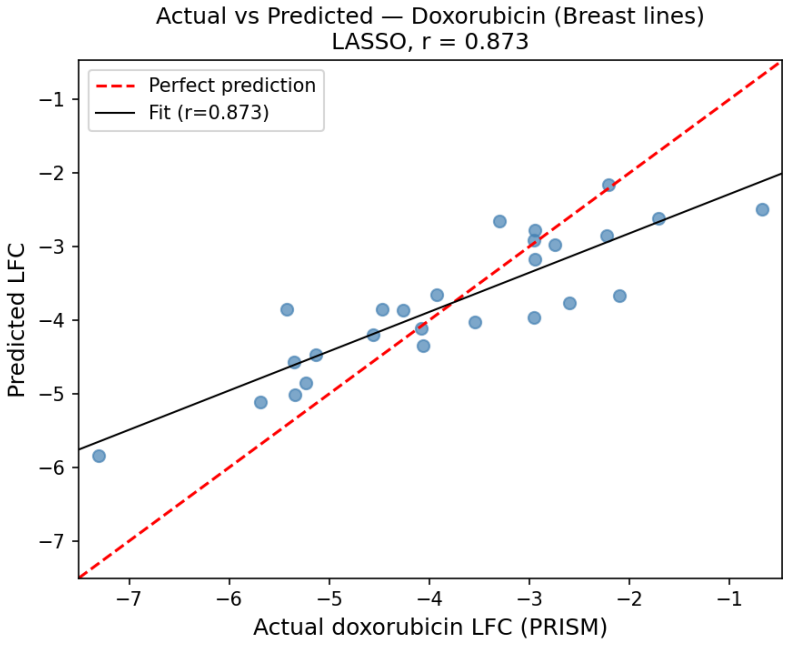


Figure 3: LASSO regression model fit for the cell line doxorubicin sensitivity signature.

This 10-gene signature was applied to the BRCA TCGA patient expression data by calculating a linear signature score, which is the dot product of normalised gene expression and LASSO coefficients. However the Cox regression of the signature score against overall survival and the Kaplan-Meir analysis using the optimal survival-stratifying threshold did not attain significance, with the Cox regression model obtaining a HR, 95% CI, p-value and a C-index of 1.17, 0.90 to 1.51, 0.238 and 0.550, respectively, whilst the Kaplan-Meir analysis produced a log-rank p value of 0.102 that can be seen in Figure 4.

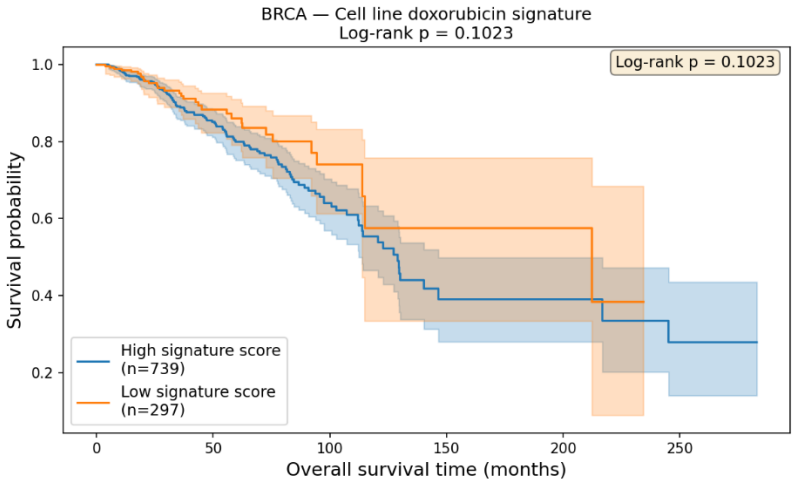


Figure 4: Cell line derived doxorubicin sensitivity signature does not stratify patient survival

One acknowledged limitation of cell-line derived drug signatures is the absence of a significant survival association. As cell lines are produced under homogenous conditions and lack the micro-environmental and stromal heterogeneity of primary tumours, a transcriptomic profile predictive of doxorubicin sensitivity in vitro does not necessarily capture the mechanisms causing varied patient outcomes in vivo.(10) The generated signature's generalisability is further limited by the small number of breast cancer cell lines within the available PRISM doxorubicin data (n=26).

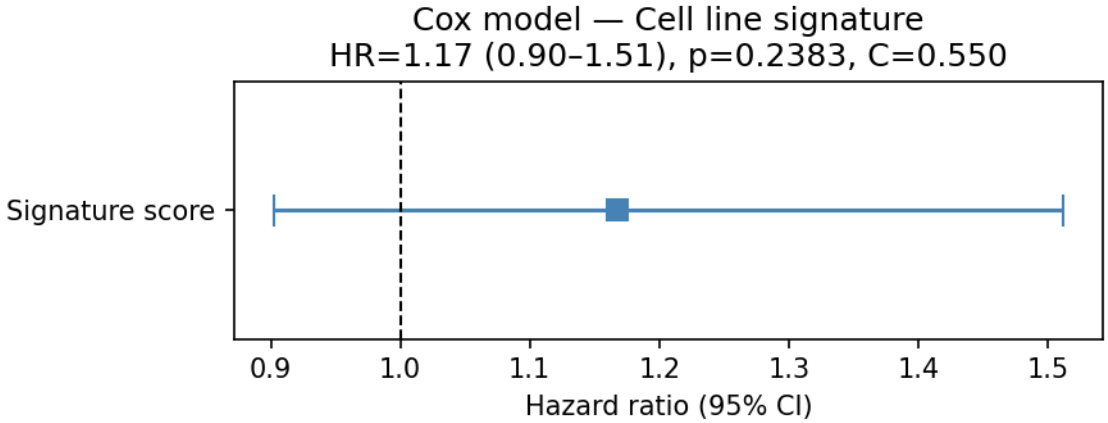


Figure 5: Cox forest plot for cell line signature

Table 3: Summary of survival associtions for ODE derived p53 phosphorylation states and cell line derived doxorubicin signature

Analysis	Optimal DDR	HR (95% CI)	Cox p	C-index	KM log-rank p	High/Low n
p53pS15	1.0	1.14 (1.01 to 1.28)	0.029	0.524	0.0008	184/834
P53pS46	0.01	0.003 (0.000 to 65.6) unstable	0.257	0.563	0.017	538/480
Cell-line	N/A	1.17	0.238	0.550	0.102	739/297

Signature		(0.90 to 1.51)				

Discussion

In the BRCA TCGA cohort, the personalised p53 ODE model identified a significant association between overall survival and simulated p53pS15 levels as it obtained a Cox p value of 0.029 and a HR of 1.14. The DDR input that proved most prognostic was the maximum dose of 1.0. This suggests that meaningful prognostic heterogeneity is captured by the degree of p53 phosphorylation under saturating DNA damage, which differs among patients because of the differences in ATM, CHK2 and WIP1 expression. Given that p53 is typically a tumour suppressor, the association's direction which is that higher p53pS15 led to worse survival, may appear contradictory. With that being said, high simulated p53pS15 at a maximum DDR level may actually indicate an over-activation of the DDR pathway in patients that have underlying signalling dysregulation. This is consistent with the finding that, in some cases, extreme p53 pathway activation may paradoxically correspond with poor outcomes.(7) It should also be noted that the hazard ratio of 1.14 that was obtained is quite modest and the C-index of 0.524 is also low, which suggests that p53pS15 alone can be seen as a weak individual predictor. Also, whilst the signal is statistically significant in a large cohort of 1,036, I think an improvement would be seen if clinical covariates such as tumour stage and subtype were also integrated.

Due to the very small magnitude of p53pS46 values at the lowest DDR input value of 0.01, p53pS46 failed to provide a significant Cox association at any of 10 DDR doses, with the best results obtained being a p value of 0.257 at a DDR does of 0.01. This Cox fit itself was in fact, numerically unstable as it achieved a HR of 0.003 and a 95% CI of 0.00 to 65.60. However, p53pS46 did reach statistical significance once patients were divided at the log-rank optimal threshold but contrasting p53pS15, higher p53pS46 was associated with better survival. Since , p53pS46 is driven by HIPK2, which is associated with the apoptotic arm of p53 activity and retained apoptotic competence would be predicted to favour outcome, this sign reversal may be biologically comprehensible.(2) Nonetheless, it is more reasonable to interpret the difference between the significant divided result and the non-significant continuous Cox result, combined with the opposite direction in regards to p53pS15, as a consequence of the threshold selection procedure inflating apparent significance rather than as solid proof of a protective effect. Therefore, p53pS46 cannot be determined as a reliable prognostic marker for this cohort.

In regard to question 2, the ODE model illustrated a trend toward predicting doxorubicin sensitivity in breast cancer cell lines as it obtained a p value of 0.074 and a r value of -0.36. This was supported with the expected directional association between p53pS15 and drug response. The small sample size of 26, which provides limited power to detect weak correlations, is the most likely cause of this trends lack of significance. The optimal DDR dose of 1.0 for patient survival differs significantly from the optimal DDR dose of 0.01 for the cell line analysis. This difference may indicate a distinction in what the model captures in each situation. At very low DNA damage input, the ODE model's output is mainly driven by

the p53 pathway's basal state, which is actually driven by the synthesis rates and the steady-state protein levels, as opposed to the ATM/CHK2 mediated phosphorylation cascade.(4) This ultimately means that the pathways capacity may be more predictive of in vitro drug sensitivity, whilst the response to maximum DNA damage may capture better, the chronic signalling differences that stratify long-term patient survival. With all of that being said, the small cell line sample size along with the modest effect sizes prevent strong conclusions on predictive usefulness in the cell line setting, though the combined Question 1 and 2 results indicate that the personalised p53 model does capture biologically meaningful variation across both patients and cell lines.

For Question 3, patient survival was not significantly predicted by the cell line-derived doxorubicin sensitivity signature that obtained a Cox p value of 0.238. Transcriptomic signatures obtained from cell lines often fail to generalise to primary tumours, which is a well-documented challenge in translational oncology.(10) This gap could be explained by a variety of factors. Firstly, the in-sample Pearson correlation r value of 0.87 most likely highlights overfitting to the training set rather than a true signal as the signature was obtained using only 26 cell lines. This small training set limits the robustness of the LASSO model regardless of any regularisation approach that was applied. Secondly, both the dosage and pharmacokinetic environment experienced by patients who undergo chemotherapy may not be accurately reflected in doxorubicin sensitivity measured at a single fixed dose in vitro.(6) Thirdly, cell lines lack the tumour microenvironment, stromal interactions along with the immune infiltration that significantly influence drug response and patient prognosis in vivo as they are grown as monolayers under uniform conditions.(10) Finally, as breast cancer is known to be molecularly heterogenous, including subtypes such as ER-positive, HER2-amplified and triple-negative all of which have distinct biology, a signature that is trained on a mixed panel of breast cancer cell lines without stratification of these subtypes may be hindered by this heterogeneity. (5)

Conclusion

When combined, these results illustrate both the benefit and limitations of personalised mechanistic modelling. Using only eight genes and a biologically based model structure, the p53 ODE approach successfully identifies a statistically significant prognostic signature via p53pS15, without the need of a large training set. In contrast to this, to generalise, the data-drive cell line approach requires sufficient cell line representation along with biological fidelity to the tumour context. Future work could focus on improving the Question 3 analysis. This could be conducted by incorporating larger cell line panels such as the GDSC2, stratifying by breast cancer subtype or even by fusing mechanistic and data-drive variables in a multivariate Cox model.

References

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